DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026 : WORKSHEET-1

Subject: Physics (086)

Chapter: 1 –Light

Date: _____ Name : _____ Class / Sec. X _____

OBJECTIVE BASED QUESTIONS

1. The laws of reflection hold true for

(a) plane mirrors only
(b) concave mirrors only
(c) convex mirrors only
(d) all reflecting surface

2. List four characteristics of the images formed by plane mirrors.

3. When a plane mirror is rotated through a certain angle, the reflected ray turns through twice as much and the size of the image:

a) is doubled
(b) is halved
(c) becomes infinite
(d) remains the same

4. If an object is placed symmetrically between two plane mirrors, inclined at an angle of 72 degrees, then the total no. of images formed is:

(a) 5
(b) 4
(c) 2
(d) infinite

5. Which statement is true for the reflection of light?

(a) The angle of incidence and reflection are equal.

(b) The reflected light is less bright than the incident light.

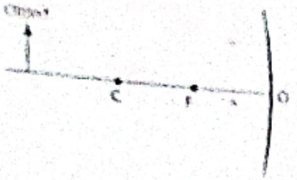
(c) The sum of the angle of incidence and reflection is always greater than 90^0.

(d) The beams of the incident light, after reflection, diverge at unequal angles.

6. The focal length of a plane mirror is

(a) 0
(b) infinite
(c) 25 cm
(d) -25 cm

7. The image shows the path of incident rays to a concave mirror.



Where would the reflected rays meet for the image formation to take place?

(a) Behind the mirror

(b) Between F and O

(c) Between C and F

(d) Beyond C

8. A beam of light incident on a plane mirror forms a real image on reflection. The incident beam is:

(a) parallel

(b) convergent

(c) divergent

(d) not certain

9. An object is placed at a distance of 40cm in front of a concave mirror of a focal length of 20 cm.

The image produced is:

(a) virtual and inverted

(b) real and erect

(c) real, inverted and of the opposite size as that of the object

(d) real, inverted and of the same size as that of the object

10. A student conducts an experiment using a convex lens. He places the object at a distance of 60 cm in front of the lens and observes that the image is formed at a distance of 30 cm behind the lens. What is the power of the lens?

(a) 0.005 dioptre

(b) 0.05 dioptre

(c) 5 dioptre

(d) 50 dioptre

11. An image of an object produced on a screen which is about 36 cm using a convex lens. The image produced is about 3 times the size of the object. What is the size of the object?

(a) 12 cm

(b) 33 cm

(c) 39 cm

(d) 108 cm

12. Image formed by a convex spherical mirror is:

(a) virtual

(c) enlarged

(b) real

(d) inverted

13. A student studies that a convex lens always forms a virtual image irrespective of its position. What causes the convex mirror to always form a virtual image?

(a) Because the reflected ray never intersects

(b) Because the reflected ray converges at a single point

(c) Because the incident ray traces its path back along the principal axis

(d) Because the incident ray of a convex mirror gets absorbed in the mirror

14. A 10 mm long awl pin is placed vertically in front pin is formed at 30cm in front of the mirror. The focal length of this mirror is:

(a) -30 cm

(c) -40 cm

(b) -20 cm

(d) -60 cm

15. Rahul conducts an experiment using an object of height of 10 cm and a concave lens with a focal length of 20 cm. The object is placed at a distance of 25 cm from the lens. Can the image be formed on a screen?

(a) Yes, as the image formed will be real

(b) Yes, as the image formed will be erect

(c) No, as the image formed will be virtual

(d) No, as the image formed will be inverted

16. Magnification produced by a rear view mirror fitted in vehicles:

(a) is less than one

(b) is more than one

(c) is equal to one

(d) can be more than or less than one, depending upon the position of the object in front of it

17. A student conducts an activity using a concave mirror with a focal length of 10 cm. He placed the object 15 cm from the mirror. Where is the image likely to form?

(a) At 6 cm behind the mirror

(c) At 6 cm in front of the mirror

(b) At 30 cm behind the mirror

(d) At 30 cm in front of the mirror

18. The image of an object placed in front of a convex mirror is formed at

(a) the object itself

(b) twice the distance of the object in front of the mirror

(c) half the distance of the object in front of the mirror

(d) behind the mirror

19. A full length of the image of a distant tall building can definitely be seen using:

(a) a concave mirror

(b) a convex mirror

(c) a plane mirror

(d) both concave as well as plane mirrors

20. A student conducts an activity using a flask of height 15 cm and a concave mirror. He finds that the image formed is 45 cm in height. What is the magnification of the image?

(a) -3 times

(b) -1/ 3 times

(c) 1/ 3 times

(d) 3 times

21. Which of the following can make a parallel beam of light from a point source incident on it?

(a) Concave mirror as well as convex lens

(b) Convex mirror as well as concave lens

(c) Two plane mirrors placed at 90degree to each other

(d) Concave mirror as well as concave lens

22. A student studies that the speed of light in air is 300000 km/sec, whereas that of speed in a glass slab is about 197000 km/sec. What causes the difference in the speed of light in these two media?

(a) Difference in density

(b) Difference in temperature

(c) Difference in the amount of light

(d) Difference in the direction of wind flow

23. Which mirror can form a real image of an object?

24. The swimming pool appears to be less deep than it actually is. Which of the following phenomena is responsible for this?

25. A 3 cm high object is placed at a distance of 80cm from a concave lens of focal length 20cm. Find the size of the image formed.

26. A convex mirror used for the rear view on an automobile has a focal length of 2.5m. A car is located at a distance of 4m from the mirror. In the above scenario the sign that should be assigned to f and u is

(a) u positive, f negative

(b) f positive, u negative

(c) f and u are positive

(d) f and u negative

27. An object is kept at a distance more than twice the focal length (F) from a concave mirror. The distance the image formed will be

(a) less then F
(b) equal to F
(c) between F and 2 F
(d) More than 2 F

28. The speed of light, in a given medium is $\frac{2}{3}$ of its speed in vacuum. The absolute refractive index of the medium equals to

(a) 9/4
(b) 4/9
(c) 3/2
(d) 2/3

29. A ray passing through the centre of curvature of a concave mirror is inclined at an angle α to its principal axis. The angle of reflection for this ray equals

(a) 0°
(b) $\frac{\alpha}{2}$
(c) α°
(d) 90°

30. Beams of light are incident through the holes C and D respectively as shown in the figure. Which of the following could be inside the box?

(a) A rectangular glass slab
(b) Convex lens
(c) Concave lens
(d) Prism

31. The power of a lens is -4.0 D. what is the nature of the lens?
(a) Plane
(b) Concave
(c) Convex
(d) Plano convex

32. Where should an object be placed in front of a convex lens to get real image of the size of the object?
(a) At focus
(b) At 2F
(c) At Infinity
(d) Between optical centre and focus.

33. In torches, search lights and head lights of vehicles the bulb is placed
(a) Between pole and focus
(b) Very near to the focus
(c) Between focus and centre of curvature
(d) At centre of curvature

34. Figure shows a ray of light as it travels from medium A to B. Refractive index of medium B with respect to A is

(a) $\frac{\sqrt{3}}{\sqrt{2}}$
(b) $\frac{1}{\sqrt{2}}$
(c) $\frac{\sqrt{2}}{\sqrt{3}}$

(d) $\frac{\sqrt{3}}{\sqrt{2}}$

35. When a ray of light goes from one medium to another, there is
(a) Always a change in its speed as well as direction
(b) No change in speed and direction
(c) A change in speed but no change in direction
(d) A change in direction but constant speed.

36. Name the mirror that can give an erect and enlarged image of an object?

37. The rays, parallel to the principal axis, of a spherical mirror, actually meet at a point 20 cm distant from its pole. Identify the mirror and give its focal length.

38. Name the kind of surfaces that
(i) Reflect
(ii) Refract most of the light falling on them.

39. Define one dioptre of power of a lens?

40. Find the power of a concave lens of focal length 2 m.

41. Which one of the following materials cannot be used to make a lens?
(a) Water
(b) Glass
(c) Plastic
(d) Clay

42. The image formed by a concave mirror is observed to be virtual, erect and larger than the object. Where should be the position of the object?
(a) Between the principal focus and the centre of curvature.
(b) At the centre of curvature
(c) Beyond the centre of curvature
(d) Between the pole of the mirror and its principal focus.

43. Where should an object be placed in front of convex lens to get a real image of the size of the object?
(a) At the principal focus of the lens.
(b) At twice the focal length
(c) At infinity
(d) Between the optical centre of the lens and its principal focus.

44. A spherical mirror and thin spherical lens have each of focal length of -15 cm. the mirror and lens are likely to be
(a) Both concave
(b) Both convex

(c) The mirror is concave and the lens is convex
(d) The mirror is convex and lens is concave.

45. No matter how far you stand from a mirror, your image appears erect. The mirror is likely to be
(a) Plane
(b) Concave
(c) Convex
(d) Either concave or convex.

46. Which of the following lens would you prefer to use while reading small letters found in a dictionary?
(a) A convex lens of focal length 50 cm
(b) A concave lens of focal length 50 cm
(c) A convex lens of focal length 5 cm
(d) A concave lens of focal length 5 cm

47. A ray AFB is incident on a spherical mirror whose centre of curvature is 2 F. In which direction will it reflect?

48. A ray of light is incident at angle of 35° to a plane surface. What will the angle of reflection?

49. How does image changes when the face is slowly moved away from inner face of a shining spoon?

50. Due to which property of light, sharp shadow of an object is obtained?

51. A person uses concave minor for shaving, where should he position his face in front of it?

52. A ray of light is incident on a concave mirror along its principal axis. What will be the angle of reflection?

53. What does negative sign in the value of magnification of a mirror indicate?

54. Which of the two has a great power? A lens of short focal length or a lens of large focal length?

55. Which mirror is used as rear view mirror in vehicles and why?

56. The size of an object is 2 cm. The magnification produced by a mirror is +1. What is the size of the image?

57. When a ray of light passes from a denser medium to a rarer medium which angle is greater: angle of incidence or angle of refraction?

58. The power of a lens is -2D. Is the lens convex or concave?

59. Focal length of a convex mirror is 10 cm. Find the radius of curvature of the mirror?